Approximate homomorphism form of Polynomial Freiman-Ruzsa

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github:teorth-pfr:4c56de547b2f33b0bc47b92f3bfbcebc6457a776;
Apache-2.0

Editorial note. Source-organized author material only. All mathematical source spans are retained unchanged. Blueprint lean/proves/uses metadata is hidden as formatting only; numbering is local to this material wrapper.

Editorial note. Ordinary cited prerequisites and concise author arguments are accepted as such. No independent mathematical certification or public ID allocation is claimed.

Editorial note. Formatting repair after actual visual inspection: xurl allows the original long author URL to wrap within the text block.

Allowed standard prerequisites (editorial).

- Standard finite Shannon entropy identities, conditional entropy and finite abelian group algebra are cited in their original Blueprint formulations.
- Named prior PFR/entropy or BSG results are allowed prerequisites; their original statements are retained where used, without demanding recursively expanded proofs.

1 Proof of PFR

Lemma 1 (Ruzsa covering lemma)**.** *If A, B are finite non-empty subsets of a group G, then A can be covered by at most $|A+B|/|B|$ translates of $B-B$.*

Proof. Cover A greedily by disjoint translates of B. □

Proposition 2. *For any random variables Y_1, Y_2, there exist a subgroup H such that*

$$2\rho(U_H) \leq \rho(Y_1) + \rho(Y_2) + 8d[Y_1; Y_2].$$

Corollary 3. *If $|A+A| \leq K|A|$, then there exists a subgroup H and $t \in G$ such that $|A \cap (H+t)| \geq K^{-4}\sqrt{|A||H|}$, and $|H|/|A| \in [K^{-8}, K^8]$.*

Corollary 4. *If $|A+A| \leq K|A|$, then there exist a subgroup H and a subset c of G with $A \subseteq c+H$, such that $|c| \leq K^5|A|^{1/2}/|H|^{1/2}$ and $|H|/|A| \in [K^{-8}, K^8]$.*

Lemma 5 (Cauchy–Schwarz bound)**.** *If G is a finite additive group and A, B are subsets of G, then*

$$\left(\frac{|\{(a,a') \in A \times A : a + a' \in B\}|}{|G|^2}\right)^2 \leq \frac{|B|}{|G|} E(A).$$

The left-hand side is the square of the density of a subset of $G \times G$, and the right-hand side involves the density of a subset of G.

Proof. If B is empty then the claim is trivial (with the Lean convention $0/0$), so without loss of generality B is non-empty. We can rewrite

$$|\{(a,a') \in A \times A : a + a' \in B\}| = \sum_{b \in B} r(b)$$

where $r : G \to \mathbb{N}$ is the counting function

$$r(b) := |\{(a,a') \in A \times A : a + a' = b\}|.$$

From double counting we have

$$\sum_{b \in G} r(b)^2 = |G|^3 E(A).$$

The Cauchy–Schwarz inequality then gives

$$\Big(\sum_{b \in B} r(b)\Big)^2 \leq |B| \sum_{b \in B} r(b)^2 \leq |B| \sum_{b \in G} r(b)^2 = |B|\,|G|^3 E(A),$$

and the claim follows on dividing both sides by $|G|^4$. □

Lemma 6 (Balog–Szemerédi–Gowers lemma)**.** *Let G be a finite abelian group, and let A be a non-empty subset of G with $E(A) \geq \frac{1}{K}\left(\frac{|A|}{|G|}\right)^3$ for some $K \geq 1$. Then there is a subset A' of A with $|A'| \geq |A|/(C_1 K^{C_2})$ and $|A' - A'| \leq C_3 K^{C_4} |A|$, where*

$$C_1 = 2^4, C_2 = 1, C_3 = 2^{10}, C_4 = 5.$$

The Lean statement gives the two conclusions as inequalities between densities; these are equivalent to the ones above, the common factor $|G|$ cancelling.

Proof. See `https://terrytao.files.wordpress.com/2024/01/simplebsg.pdf`. □

2 Homomorphism version of PFR

In this section, G, G' are finite abelian 2-groups.

Lemma 7 (Hahn-Banach type theorem)**.** *Let H_0 be a subgroup of G. Then every homomorphism $\phi : H_0 \to G'$ can be extended to a homomorphism $\tilde{\phi} : G \to G'$.*

Proof. By induction it suffices to treat the case where H_0 has index 2 in G, but then the extension can be constructed by hand. □

Lemma 8 (Goursat type theorem)**.** *Let H be a subgroup of $G \times G'$. Then there exists a subgroup H_0 of G, a subgroup H_1 of G', and a homomorphism $\phi : G \to G'$ such that*

$$H := \{(x, \phi(x) + y) : x \in H_0, y \in H_1\}.$$

In particular, $|H| = |H_0||H_1|$.

Proof. We can take H_0 to be the projection of H to G, and H_1 to be the slice $H_1 := \{y : (0, y) \in H\}$. One can construct ϕ on H_0 one generator at a time by the greedy algorithm, and then extend to G by Theorem 7. The cardinality bound is clear from direct counting. □

Theorem 9 (Approximate homomorphism form of PFR)**.** *Let G, G' be finite abelian 2-groups. Let $f : G \to G'$ be a function, and suppose that there are at least $|G|^2/K$ pairs $(x, y) \in G^2$ such that*

$$f(x + y) = f(x) + f(y).$$

Then there exists a homomorphism $\phi : G \to G'$ and a constant $c \in G'$ such that $f(x) = \phi(x) + c$ for at least $|G|/(2^{144} * K^{122})$ values of $x \in G$.

Proof. Consider the graph $A \subset G \times G'$ defined by

$$A := \{(x, f(x)) : x \in G\}.$$

Clearly, $|A| = |G|$. By hypothesis, we have $a + a' \in A$ for at least $|A|^2/K$ pairs $(a, a') \in A^2$. By Theorem 5, this implies that $E(A) \geq \frac{1}{K^2}\left(\frac{|A|}{|G \times G'|}\right)^3$. Applying Theorem 6, we conclude that there exists a subset $A' \subset A$ with $|A'| \geq |A|/C_1K^{2C_2}$ and $|A' + A'| \leq C_1C_3K^{2(C_2+C_4)}|A'|$. Applying Theorem 4, we may find a subspace $H \subset G \times G'$ such that $|H|/|A'| \in [L^{-8}, L^8]$ and a subset c of cardinality at most $L^5|A'|^{1/2}/|H|^{1/2}$ such that $A' \subseteq c + H$, where $L = C_1C_3K^{2(C_2+C_4)}$. If we let H_0, H_1 be as in Theorem 8, this implies on taking projections the projection of A' to G is covered by at most $|c|$ translates of H_0. This implies that

$$|c||H_0| \geq |A'|;$$

since $|H_0||H_1| = |H|$, we conclude that

$$|H_1| \leq |c||H|/|A'|.$$

By hypothesis, A' is covered by at most $|c|$ translates of H, and hence by at most $|c||H_1|$ translates of $\{(x, \phi(x)) : x \in G\}$. As ϕ is a homomorphism, each such translate can be written in the form $\{(x, \phi(x) + c) : x \in G\}$ for some $c \in G'$. The number of translates is bounded by

$$|c|^2\frac{|H|}{|A'|} \leq \left(L^5\frac{|A'|^{1/2}}{|H|^{1/2}}\right)^2\frac{|H|}{|A'|} = L^{10}.$$

By the pigeonhole principle, one of these translates must then contain at least $|A'|/L^{10} \geq |G|/(C_1C_3K^{2(C_2+C_4)})^{10}(C_1K^{2C_2})$ elements of A' (and hence of A), and the claim follows. □